

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (Bsc.IT Semester III)**  
**Data Structures Practical**

**Practical - IV**

|                                |                                     |
|--------------------------------|-------------------------------------|
| Roll No. :S040                 | Name:Aditya Sopte                   |
| Class :SY BSC IT               | Batch : B2                          |
| Date of Assignment :15-07-2026 | Date/Time of Submission :15-07-2026 |

Queue Application:

- a. Write a program to: a. Implement a queue using an array.

Code:

```
from array import *

que = array('i', [])

def enqueue(data):
    que.append(data)
    print("Data inserted:", data)

def dequeue():
    if len(que) == 0:
        print("Queue is empty")
    else:
        print("Data removed:", que.pop(0))

def peek():
    if len(que) == 0:
        print("Queue is empty")
    else:
        print("First element is:", que[0])

def show():
    print("Queue:", que)
    print("Aditya Sopte S040")

enqueue(30)
enqueue(70)
enqueue(60)
show()
peek()
dequeue()
show()
```

Output:

```
= RESTART: C:/Users/IT-23/AppDa
Data inserted: 30
Data inserted: 70
Data inserted: 60
Queue: array('i', [30, 70, 60])
Aditya Sopte S040
First element is: 30
Data removed: 30
Queue: array('i', [70, 60])
Aditya Sopte S040
```

- b. Simulate a simple queuing system (e.g., customer service queue).

Code:

```
*pr 4.py - C:/Users/IT-23/AppData/Local/Programs/Python/Python314/data s/pr 4.py (3.14.6)*
File Edit Format Run Options Window Help

import time

que = [("C1",2), ("C2",5), ("C3",7), ("C4",0)]

print(" ----- Customer Order ----- ")
print(que)

while que:
    Order, timer = que.pop(0)
    print(f" Serving: {Order} {timer} sec")
    time.sleep(timer)

print(que)

print("Aditya sopte S040")
```

Output:

```
= RESTART: C:/Users/IT-23/AppData/Local/Progr
----- Customer Order -----
[('C1', 2), ('C2', 5), ('C3', 7), ('C4', 0)]
Serving: C1 2 sec
Serving: C2 5 sec
Serving: C3 7 sec
Serving: C4 0 sec
[]
Aditya sopte S040
```

### Curiosity Questions:

1. Where do you see queues in daily life?
2. Why is **FIFO (First In First Out)** fair in real life? Can you think of situations where it's *not* fair?
3. What will happen if you try to **enqueue more elements than the queue size**?
4. What if you **dequeue from an empty queue**?
5. Imagine you're designing a **hospital emergency system**. Should it use a simple queue or something else? (hint: priority queue).
6. If **5 customers arrive at the same time**, but the service desk can serve only 2 at once, how would you design that?